

Ekjot Brar

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Education

University of Alberta - *Bachelor of Applied Science*

September 2023- April 2028

- Major in Electrical Engineering; Minor in Applied Mathematics
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Career Experience

Nokia - *Service Router Platform Test Dev*

January 2026 - April 2026

- Engineered and scaled modular SR/SR-L router topologies, improving reusability and efficiency of network validation frameworks.
- Developed Python-based automation pipelines in a Linux (Ubuntu) regression environment for carrier-grade router hardware/software validation.
- Built dynamic Precision Time Protocol (PTP) test environments with parameterized configurations (IP, VLAN, interface speeds) to accelerate automated test setup.
- Conducted system-level validation of synchronization technologies (PTP, GNSS, SyncE), ensuring accurate timing and clock alignment across network devices.
- Designed object-oriented Python abstractions mapped to YANG data models for real-time configuration and monitoring of central frequency clock systems.

Tetra Tech - *CAN. Technologist*

May 2025 - August 2025

- Supported field deployment and troubleshooting of advanced pavement inspection systems (LiDAR, LCMS, laser sensors) ensuring reliable high-resolution data collection.
 - Developed a QR-based tracking system for hard drives using structured database workflows to improve data traceability and inventory management.
 - Processed and validated pavement datasets using MATLAB, identifying anomalies (bridges, bumps) and improving data quality assurance workflows.
 - Developed C/C++-based tools for configuration and data handling of imaging systems (Ladybug camera), supporting integration into analysis pipelines.
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Extra-Curricular Experience

ECO Car, University of Alberta - *Electrical Team Member*

May 2025 - PRESENT

- Contributed to Shell Eco-Marathon hydrogen vehicle development, gaining hands-on experience with embedded systems and low-power electronics relevant to semiconductor applications.
- Designed and routed a 2-layer PCB in KiCad, including schematic capture, component selection, and layout optimization for efficient electronic system integration.
- Programmed an STM32 microcontroller for real-time control and PWM-based LED operation, building embedded C and digital control experience.
- Assembled and soldered SMD/through-hole components, supporting hardware prototyping and circuit validation.
- Implemented FPGA digital logic using VHDL, including finite state machines and combinational/sequential circuit design.

AlbertaSat, University of Alberta - *Iris Deputy Lead*

September 2022 - September 2025

- Contributed to the development and microgravity testing of a gecko-adhesive debris capture system (CAN-RGX), focusing on system-level performance validation under dynamic and structural constraints
- Performed vibrational and structural analysis of CubeSat subsystems using ANSYS Mechanical and MATLAB, evaluating modal response, stress distribution, and mechanical stability of critical components
- Supported Ex-Altia 3 CubeSat hardware development, including hinge mechanism design, CAD modeling (SolidWorks, Onshape, AutoCAD), and investigation of UHF integration with emphasis on mechanical-electrical system reliability